Fantasy Basketball Roster Analytics

Software Requirements Document

Course: CISC 3171 | **Instructor:** Alec Malstrom | **Term:** Fall 2026

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Status: Draft

1. Purpose

The Predictive NBA Trade & Roster Evaluator is a web app that acts like a "weather forecast" for fantasy basketball. Instead of looking backward at past stats like Yahoo or ESPN, it predicts how many fantasy points your team will score in the future based on upcoming schedules, team changes, and player injuries.

2. Scope

2.1 In scope

Evaluating trades up to 4-for-4 players.
Projections for 1-week, 2-week, and rest-of-season windows.
Player search, custom league scoring, and roster saving.
Daily optimal lineup recommendations.
Ranked add/drop waiver wire recommendations.
Daily automated stat refreshes with timestamps.

2.2 Out of scope

Direct league imports from ESPN, Yahoo, or Sleeper.
Mobile push notifications or SMS alerts.
Native iOS or Android apps (web browser only).
Sports gambling, betting odds, or DFS salary cap tools.

3. Context

3.1 User types

Casual Manager: Checks a few times a week. Wants simple "Helps" or "Hurts" answers without math.
Competitive Manager: Checks daily. Wants projection ranges, confidence levels, and full daily breakdowns.

3.2 Operating environment

Desktop and mobile web browsers (375px phone screen up to desktop).
Daily updated NBA statistical data and schedules.

3.3 Assumptions

Users enter their rosters manually during setup.
Users know their league's scoring system or use default points settings.

4. Key Features

ID	Feature	Description	Priority	Stories
F1	Roster Setup	Search players, save roster, set custom scoring	P1	3

F2	Trade Evaluator	Input trades, calculate net FP, view verdict & top factors	P1	3
F3	Projection Engine	Calculate schedule density, B2Bs, and injuries	P1	3
F4	Daily Lineup	Show daily projections and recommend starting lineups	P2	2
F5	Waiver Wire	Rank top add/drop moves by projected 7-day gain	P2	2
F6	Data Refresh	Daily updates, timestamp display, and error fallback	P1	2

5. User Stories

F1: Roster Setup

S1.1: Building a roster
As a manager, I want to search players by name, so I can build my team.
Acceptance Criteria: Searching shows active NBA players. Adding duplicate players displays an error.
S1.2: Customizing scoring
As a manager, I want to edit scoring weights, so projections match my league.
Acceptance Criteria: Applies default points rules on setup. Editing point values instantly recalculates projections.
S1.3: Saving roster
As a manager, I want an account, so my roster stays saved.
Acceptance Criteria: Saved rosters persist across sessions when logged in.

F2: Trade Evaluator

S2.1: Evaluating a trade
As a manager, I want to evaluate a trade proposal, so I can see net point changes.

Acceptance Criteria: Handles up to 4 players per side. Uneven trades factor in the best available waiver pickup for open spots.

S2.2: Understanding verdicts

As a manager, I want clear trade verdicts, so I know whether to accept.

Acceptance Criteria: Displays **Helps Your Team**, **Hurts Your Team**, or **Roughly Even**. Differences within ± 5 FP are labeled **Roughly Even**. Shows icon + label (never color alone) and top 3 reasons marked + or -.

S2.3: Seeing confidence & details

As a manager, I want to see confidence and breakdown charts.

Acceptance Criteria: Shows likely range (e.g., +6 to +29 FP) and confidence level (Low/Med/High).

F3: Projection Engine

S3.1: Changing windows

As a manager, I want 1-wk, 2-wk, and ROS windows, so I can plan ahead.

Acceptance Criteria: Toggling recalculates points for that specific schedule window.

S3.2: Injury adjustments

As a manager, I want projections to reflect injuries, so evaluations stay accurate.

Acceptance Criteria: Players marked OUT drop to 0 FP for missed games. Sidelined teammates boost remaining players' usage rates.

S3.3: Handling zero games

As a manager, I want clear messages when a team has no games.

Acceptance Criteria: Shows "No games in this window" when zero games are scheduled.

F4: Daily Lineup

S4.1: Viewing daily projections

As a manager, I want to see daily match-ups, so I can pick starters.

Acceptance Criteria: Displays slot (PG, SG, C, etc.), opponent, status, and projected daily FP.

S4.2: Optimizing lineup

As a manager, I want 1-click lineup optimization, so I don't bench points.

Acceptance Criteria: "Optimize" button swaps players to maximize daily points based on eligibility.

F5: Waiver Wire

S5.1: Add/Drop suggestions
As a manager, I want ranked waiver recommendations so I can improve my team.
Acceptance Criteria: Ranks Add/Drop pairs by projected 7-day FP gain with a 1-line explanation.
S5.2: Position filtering
As a manager, I want to filter waivers by position so I can fill weak slots.
Acceptance Criteria: Filtering restricts suggestions to selected position eligibility.

F6: Data Refresh

S6.1: Data timestamp
As a manager, I want to see when data was last updated, so I know it's fresh.
Acceptance Criteria: Displays "Data last updated" timestamp on all projection screens.
S6.2: Offline fallback
As a manager, I want the app to work during source downtime.
Acceptance Criteria: Failed refreshes display cached data alongside a warning banner.

6. Non-Functional Requirements

ID	Requirement	Rule
NFR1	**Performance**	Trade evaluation results return within 3 seconds.
NFR2	**Usability**	First-time users can run a trade eval in under 2 minutes.
NFR3	**Responsiveness**	Mobile-friendly design down to 375px width.

NFR4	**Security**	Passwords hashed, HTTPS forced, no platform logins requested.
NFR5	**Accessibility**	Verdicts use text + icons ($\mathbf{\Delta}/\mathbf{\nabla}/\approx$), never color alone.

7. Open Questions

OQ1: How should 9-category trade verdicts be summarized?

Points leagues use a single net point number. Category leagues win/lose 9 stats. We need to decide if 9-cat trades show "+2 Categories" or category-by-category wins.

Waiting on: User testing feedback.
Decided by: Product Owner.
Blocks: F2.

8. Future Directions

FD1: Automatic roster imports from Yahoo/ESPN APIs.
FD2: SMS / Push notification alerts for player injuries.

9. References

Software Pitch: Fantasy Basketball Roster Analytics Pitch Document (Group 6).
Domain Rules: Appendix A.

Appendix A: Domain Knowledge & Rules

Standard Points Scoring Weights

Points (PTS) = +1.0
Rebounds (REB) = +1.2
Assists (AST) = +1.5
Steals (STL) = +3.0
Blocks (BLK) = +3.0
Turnovers (TO) = -1.0